

RAY SWAN

GENAI / ML ARCHITECT | AI PLATFORM ARCHITECT | MLOPS / LLMOPS

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PROFESSIONAL SUMMARY

GenAI/ML Architect and hands-on AI Platform Architect with 25 years delivering enterprise software, data, integration, and transformation programs. Builds governed AI systems from infrastructure through model and agent runtime: Azure ML, Azure Databricks, Microsoft Foundry, MLflow, Terraform, GitHub Actions, Entra/OIDC, RAG/vector retrieval, and production lifecycle automation. Combines architecture, implementation, product leadership, and evidence-based governance across Azure, GCP, and AWS.

ARCHITECTURE & ENGINEERING EXPERTISE

AI Platforms & GenAI: Azure ML; Azure Databricks; Microsoft Foundry; MLflow; RAG; vector retrieval; grounded assistants; agentic systems; MCP-oriented tool boundaries; model evaluation

MLOps / LLMOps: Local-to-cloud lifecycle; champion/challenger; model registry; promotion/retain gates; drift monitoring; retraining; rollback-ready versioning; observability-ready contracts

Cloud & IaC: Terraform; GitHub Actions; Entra ID/OIDC; RBAC; Docker; Azure; GCP/Vertex AI; AWS/SageMaker; APIs; microservices; PostgreSQL

Governance & Security: Digest-bound saved plans; least-privilege identity separation; immutable evidence/provenance; state-machine guardrails; deterministic validation; bounded permissions

SELECTED PRODUCT ARCHITECTURE

Tokyo Eye — Scientific AI / Drug Discovery | Founder & Technical Lead | Eidetix Bio Research | 2025–Present

- Architected a governed scientific-AI platform using hyperbolic GNN and atom-level Transformer/Mixture-of-Experts architectures for structural inference and ligand-protein binding prediction.
- Pivoted from an underperforming PyTorch GNN to a replacement atom-level Transformer architecture; designed, trained, and evaluated it in one day, producing same-day Pearson correlation of 0.407.
- Built MLflow governance with preregistered estimates, immutable metric gates, ablations, provenance, and preservation of failed experiments; scientist-facing LLM workflows use XState guardrails, Python Hypothesis validation, deterministic structural-biology checks, and RCSB semantic search.

ADDITIONAL AI / ML SYSTEMS

Azure GenAI/ML Ops Factory — Architect & Hands-on Engineer | 2026

- Public proof: github.com/rubyrayjuntos/AI-ML-SCAFFOLD. Re-engineered Microsoft's Azure MLOps v2 Accelerator into a factory that generates modular, self-contained Azure ML project repositories with governed infrastructure.
- Terraform-managed Azure ML with keyless storage, Key Vault, Log Analytics, Entra/OIDC GitHub CI, and evidence-bound plan/apply; local-first lifecycle (prepare → train → evaluate → register) with a live MLflow run registered in Dev.

NeuroNote — Governed Agent Runtime | 2025

- Designed a dual-kernel AI governance architecture that separates generative reasoning from controlled runtime state and deterministic execution, limiting agent authority while preserving useful interaction.

Kitchen Kontrol — Independent Production Operations Product

- Independent operations product that ran in production across ~23 Sodexo school cafeterias: HACCP evidence, structured audit logging, and gloved/bilingual mid-service workflows. Public GitHub proof is [kk-logger](#) (milk/HACCP inventory logger) and [MyKitchen](#); the full Azure codebase is not public.

ENTERPRISE EXPERIENCE

Founder & Technical Lead | EIDETIX BIO RESEARCH | Oct 2025–Present

- Architected Tokyo Eye, a governed scientific-AI platform for computational drug discovery using hyperbolic GNN and atom-level Transformer/Mixture-of-Experts architectures for ligand-protein binding prediction. Own architecture, data pipeline, model training, and deployment with MLflow governance.

Operations Applications Developer / AI products | SODEXO | Aug 2025–Present

- Secondary operations role (Area Supervisor). Translates frontline operational problems into deployable AI-enabled products, combining product ownership, application development, integration design, and adoption. Kitchen Kontrol is an independent product that ran across ~23 school cafeterias; public GitHub proof is [kk-logger](#).

Technical Business Analyst & Project Manager | INTERNATIONAL REAL ESTATE SERVICES | Jan 2024–Jan 2025

- Led CRM, analytics, commercial-loan origination, API integration, ETL, process mapping, roadmap planning, and acceptance criteria for audit-ready enterprise workflows.

Senior Enterprise Program / Product / Architecture Leadership | Theragun · Sunrun · Mazda · Fox Sports · agencies | 1999–2024

- Led application modernization, CRM, e-commerce, data, integration, and platform programs for Slumberland Furniture, Sunrun, Mazda North American Operations, Fox Sports, Theragun, Lakeshore Learning, Kayne Anderson Rudnick, and HomePlus Mortgage.
- Directed cross-functional teams, platform/vendor evaluation, source-of-truth design, governance, rollout planning, and production adoption in complex Fortune 500 environments.

EDUCATION

M.A., Humanities — California State University, Dominguez Hills | B.A., Economics — University of California, Irvine